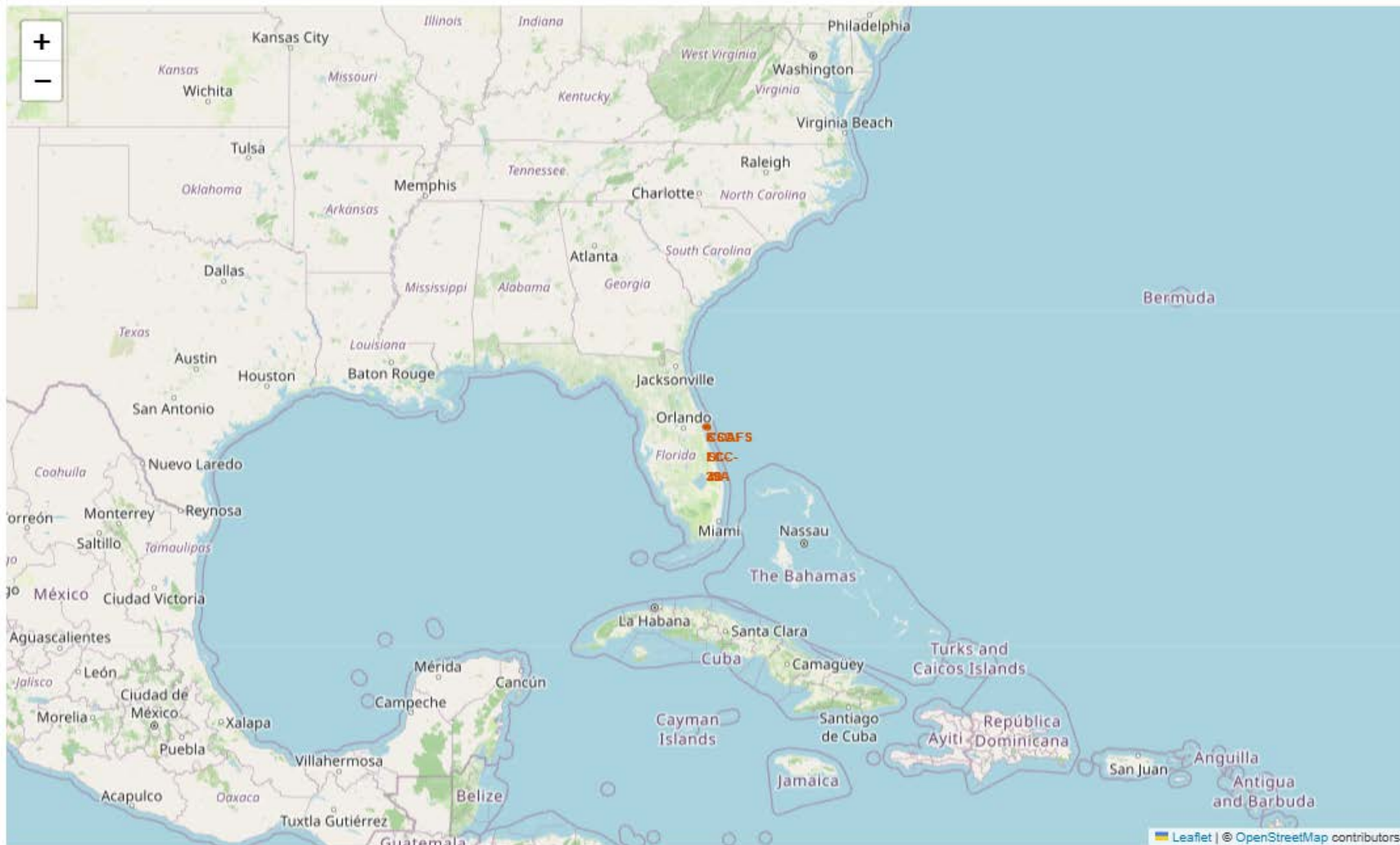


site_map

[5]:

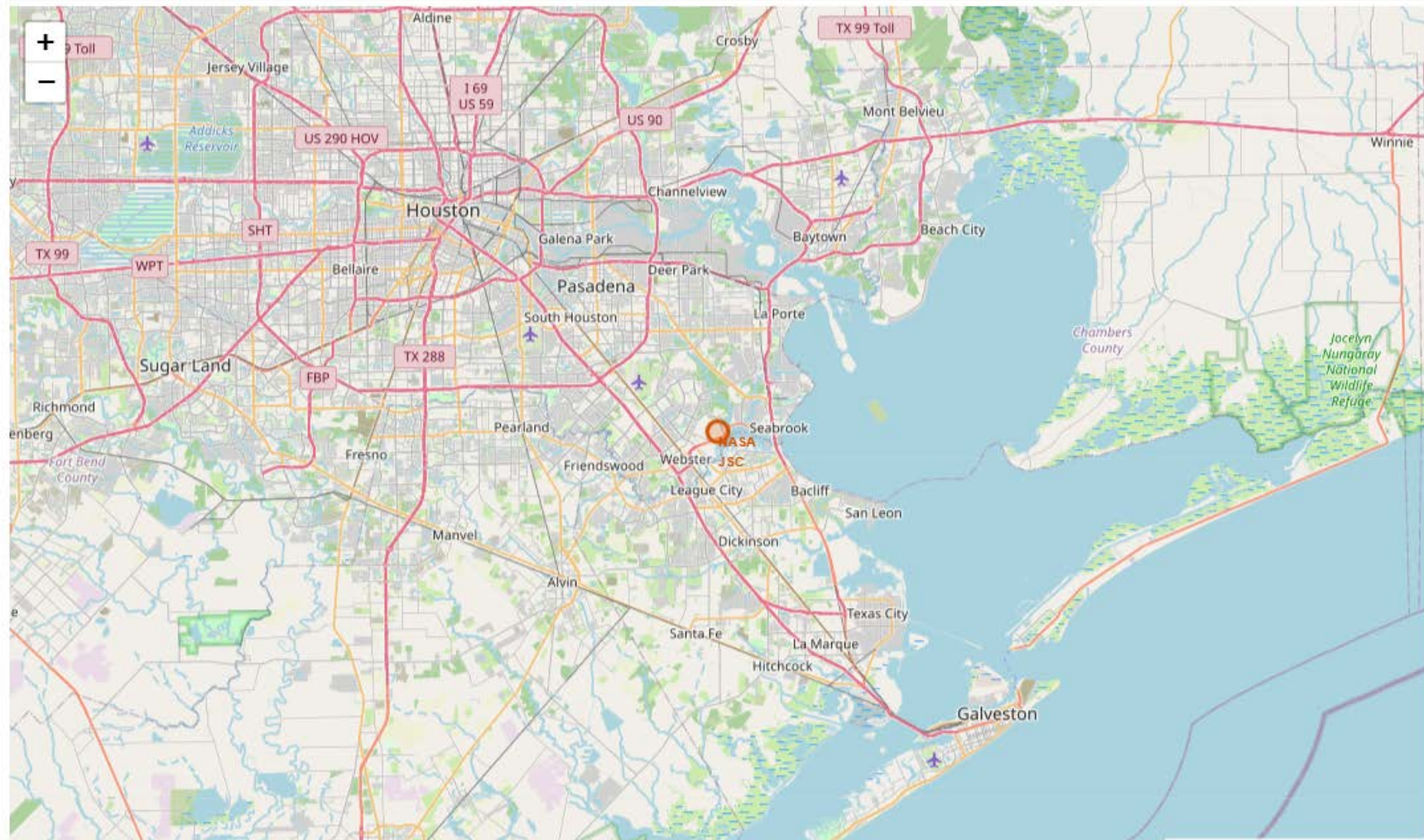


First, let's try to add each site's location on a map using site's latitude and longitude coordinates

The following dataset with the name `spacex_launch_geo.csv` is an augmented dataset with latitude and longitude added for each site.

```
icon_anchor=(0,0),  
html='<div style="font-size: 12; color:#d35400;"><b>%s</b></div>' % 'NASA JSC',  
)  
)  
site_map.add_child(circle)  
site_map.add_child(marker)
```

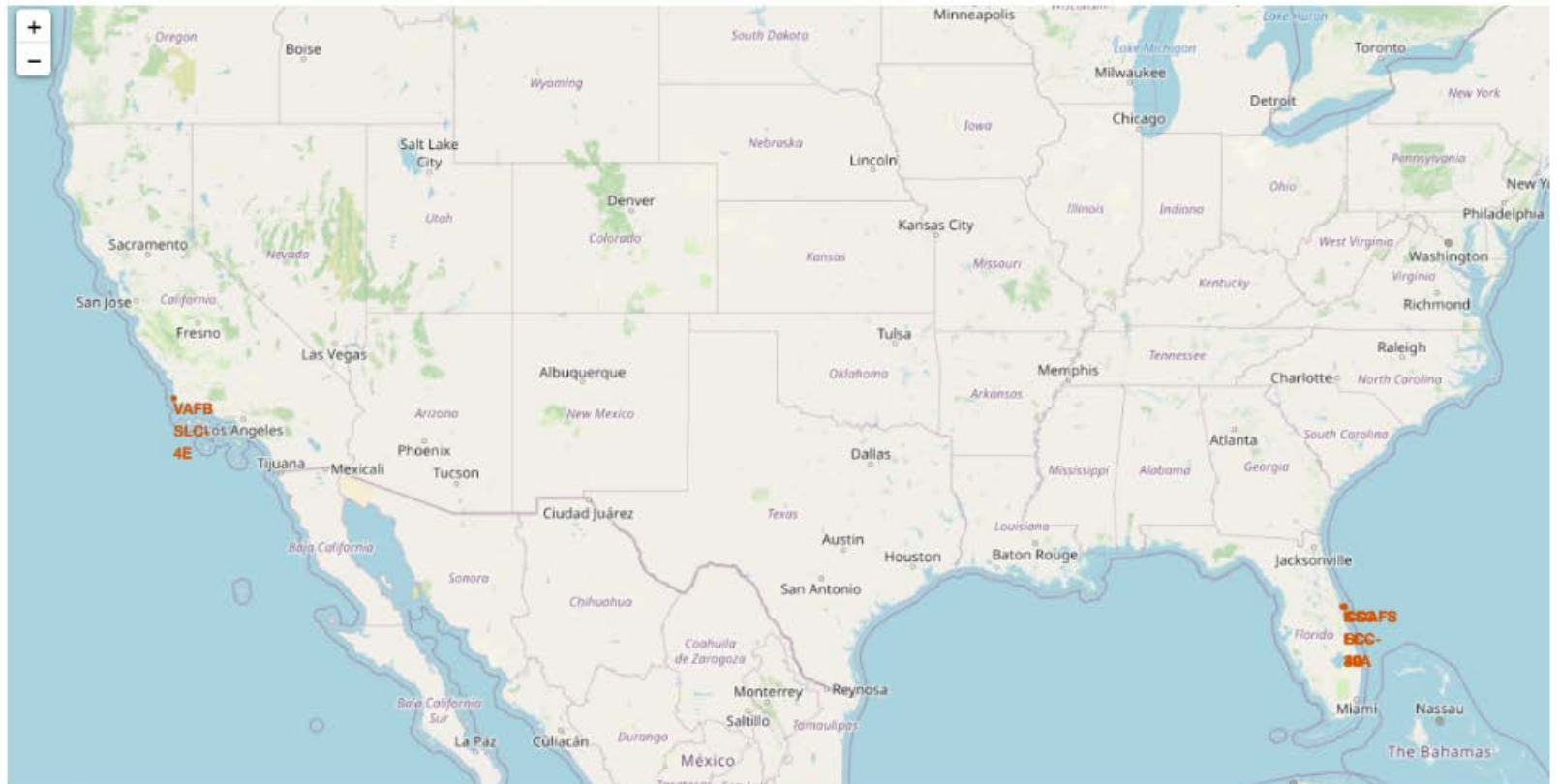
[1]:




```
folium.map.Marker(coordinate, icon=DivIcon(icon_size=(20,20),icon_anchor=(0,0), html='<div style="font-size: 12; color:#d35400;">
<b>%s</b></div>' % 'label', ))
```

```
[ ]: # Initial the map
site_map = folium.Map(location=nasa_coordinate, zoom_start=5)
# For each launch site, add a Circle object based on its coordinate (Lat, Long) values. In addition, add Launch site name as a popup Label
```

The generated map with marked launch sites should look similar to the following:

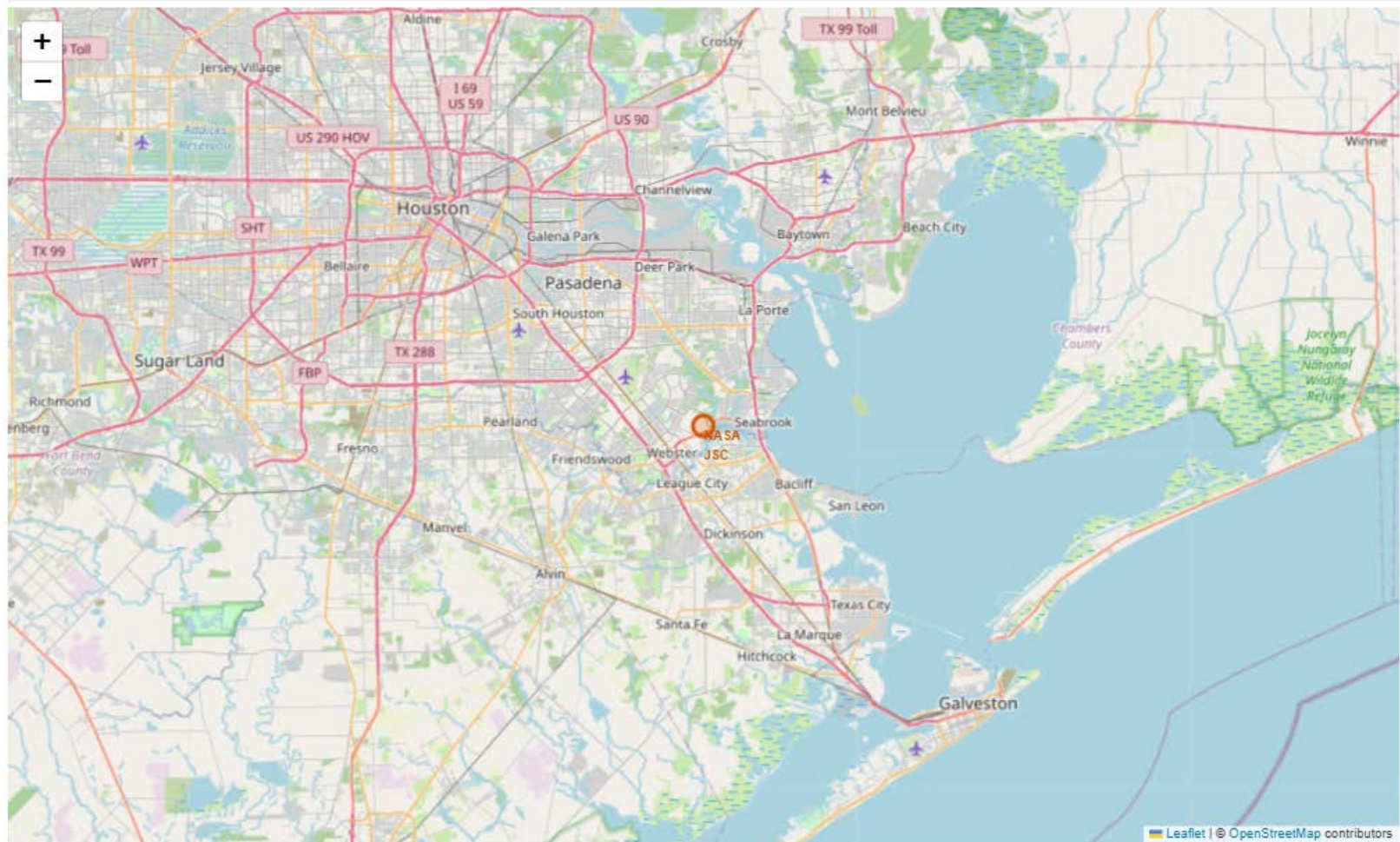


Now, you can explore the map by zoom-in/out the marked areas , and try to answer the following questions:

```
)  
marker_cluster.add_child(marker)
```

```
# Display the map with clustered markers  
site_map
```

12]:



TODO: Create a new column in `spacex_df` dataframe called `marker_color` to store the marker colors based on the `class` value

[]:

```
# Apply a function to check the value of `class` column
```

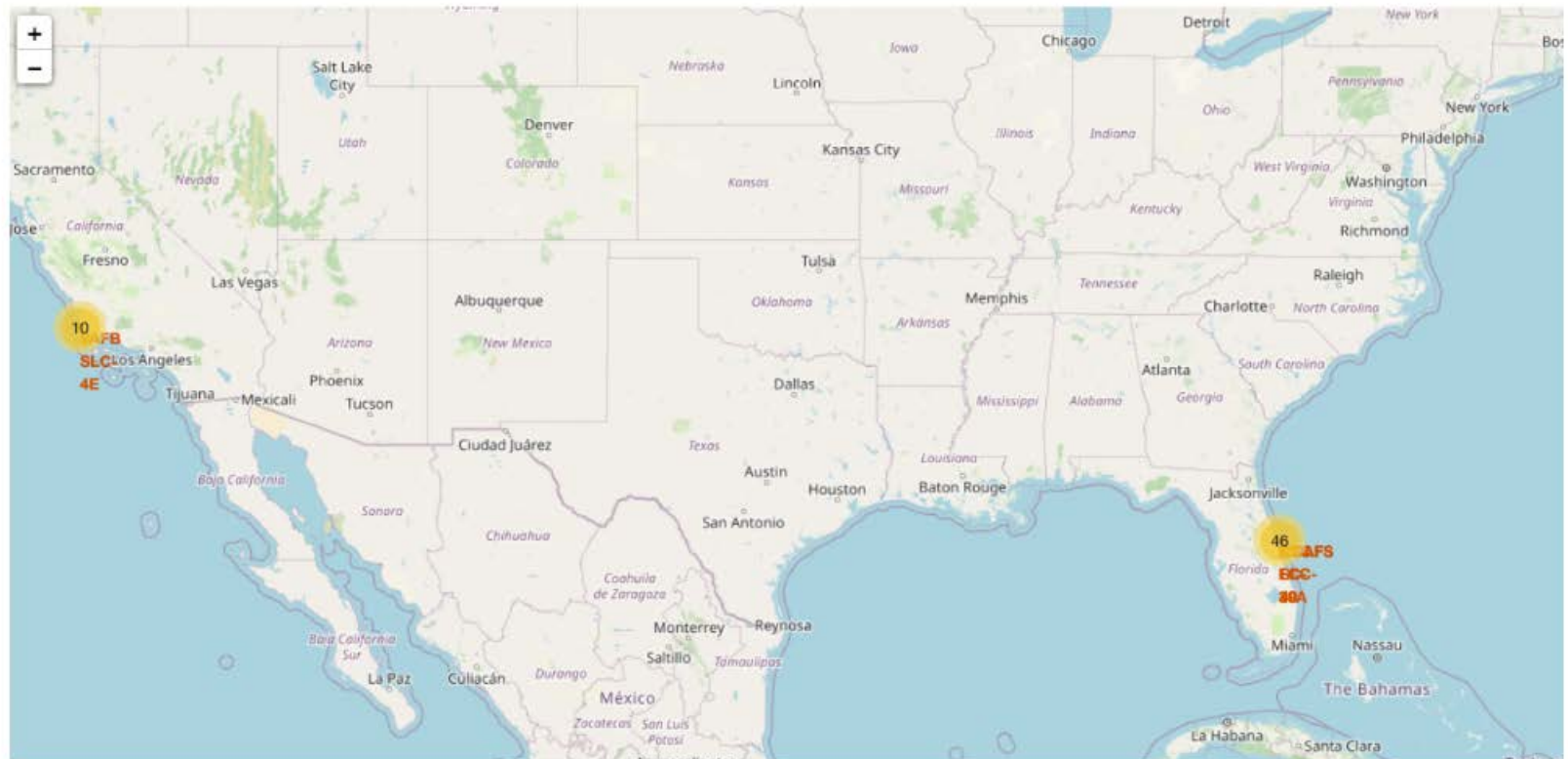
```

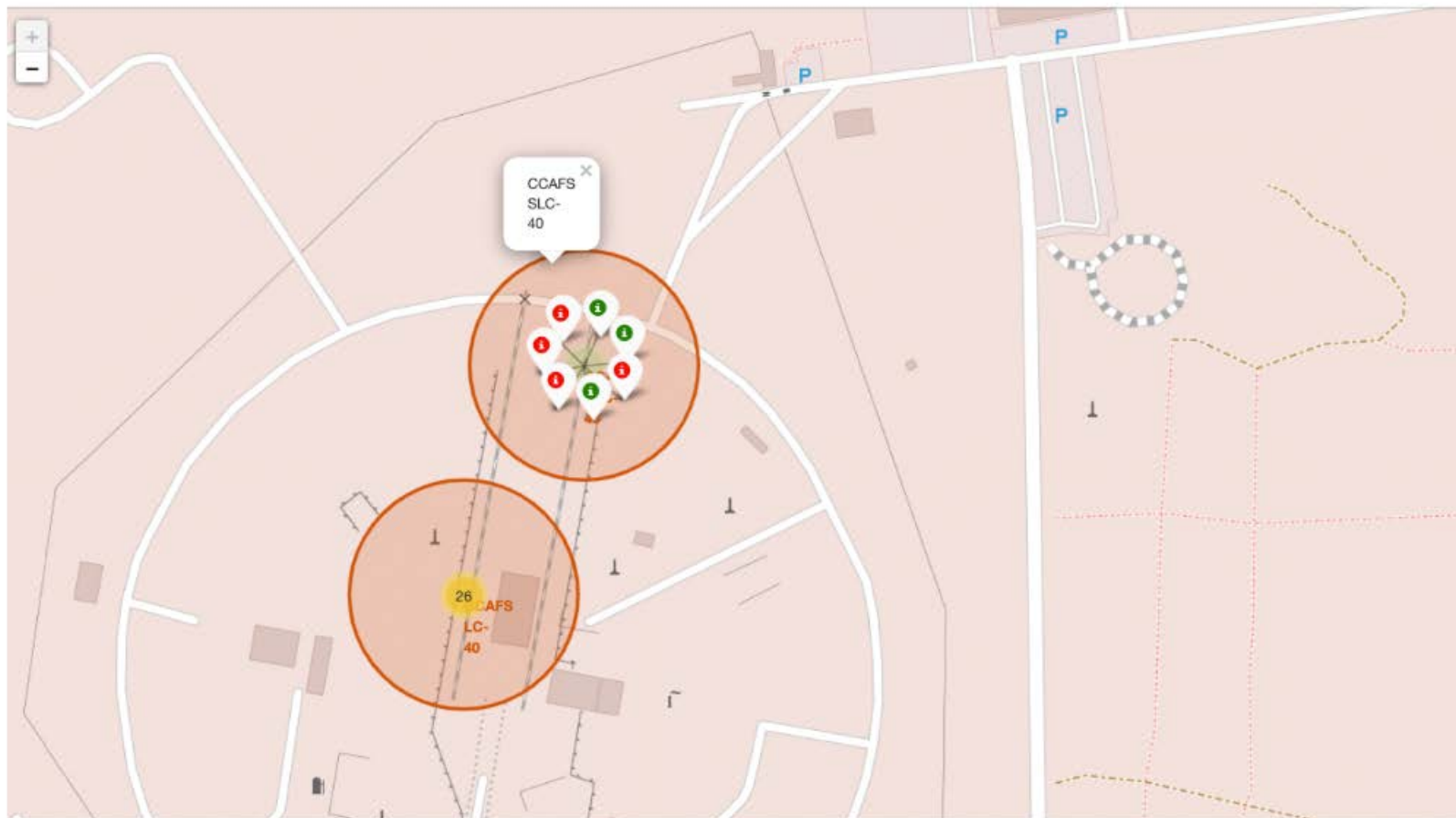
# create a Marker object with its coordinate
# and customize the Marker's icon property to indicate if this launch was succeeded or failed,
# e.g., icon=folium.Icon(color='white', icon_color=row['marker_color'])
for index, record in spacex_df.iterrows():
    # TODO: Create and add a Marker cluster to the site map
    # marker = folium.Marker(...)
    marker_cluster.add_child(marker)

```

site_map

Your updated map may look like the following screenshots:



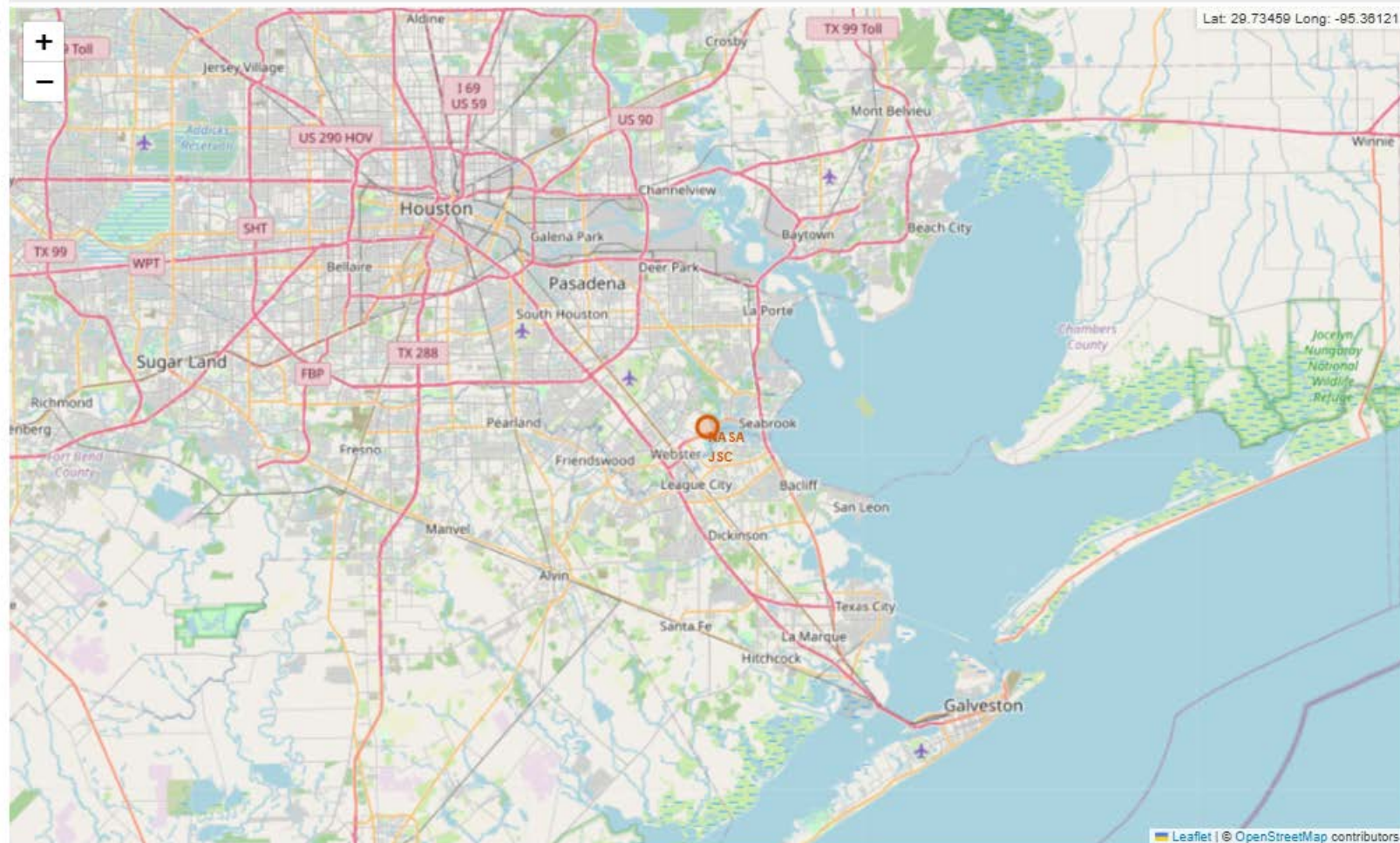


From the color-labeled markers in marker clusters, you should be able to easily identify which launch sites have relatively high success rates.

```
[ ]: # TASK 3: Calculate the distances between a launch site to its proximities
```

```
site_map.add_child(mouse_position)
site_map
```

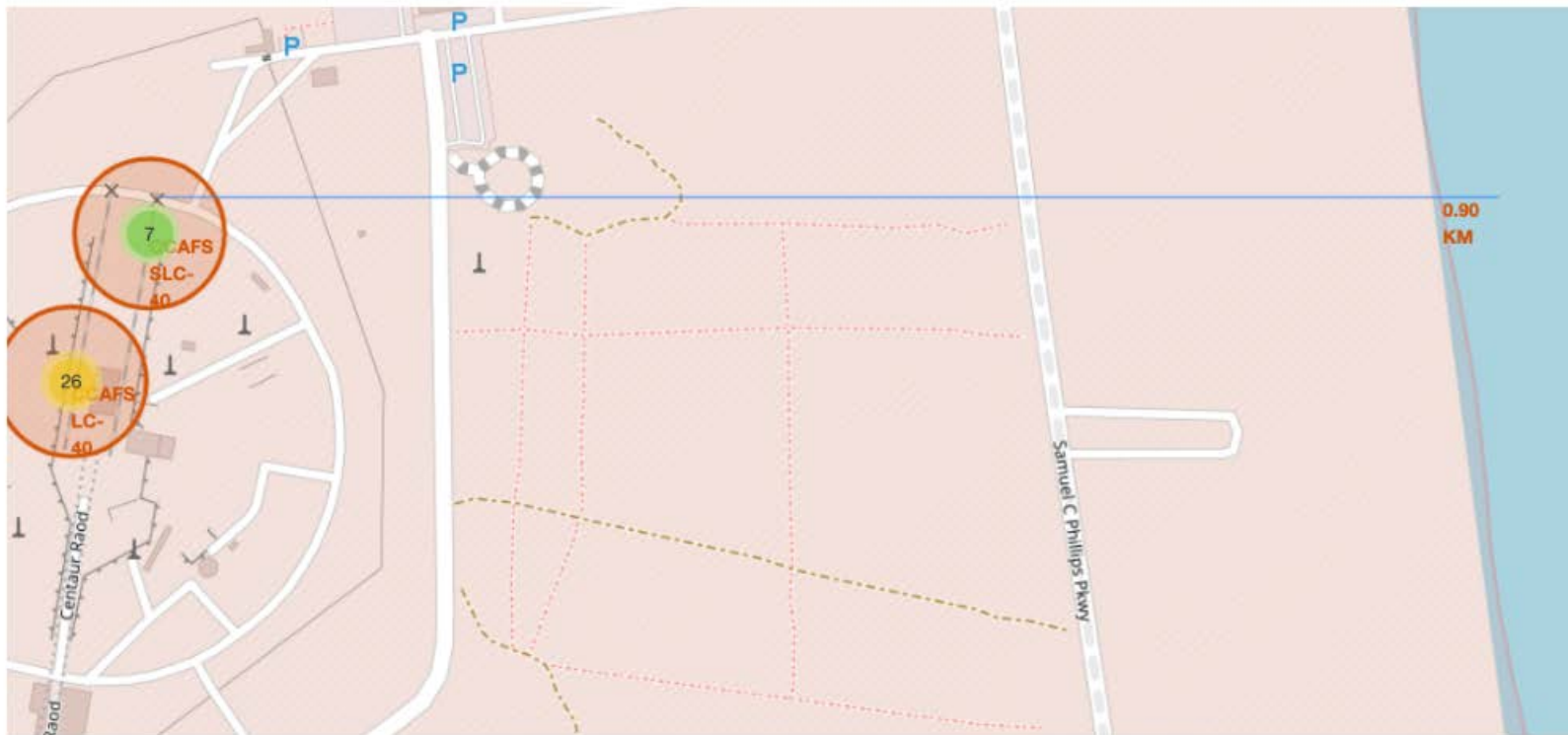
[13]:



Now zoom in to a launch site and explore its proximity to see if you can easily find any railway, highway, coastline, etc. Move your mouse to these points and mark down their coordinates (shown on the top-left) in order to the distance to the launch site.

```
[ ]: # Create a `folium.PolyLine` object using the coastline coordinates and launch site coordinate
# lines=folium.PolyLine(locations=coordinates, weight=1)
site_map.add_child(lines)
```

Your updated map with distance line should look like the following screenshot:



TODO: Similarly, you can draw a line between a launch site to its closest city, railway, highway, etc. You need to use `MousePosition` to find the their coordinates on the map first

[5]:

```
import pandas as pd
import folium
from folium.features import DivIcon

# 1. Download and read the dataset (if spacex_df is not already loaded)
spacex_df = pd.read_csv('https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/datasets/spacex_launch_geo.csv')

# 2. Select distinct launch sites with their average coordinates
launch_sites_df = spacex_df[['Launch Site', 'Lat', 'Long']].groupby(['Launch Site'], as_index=False).first()

# 3. Create the folium map
site_map = folium.Map(location=[28.562302, -80.577356], zoom_start=5)

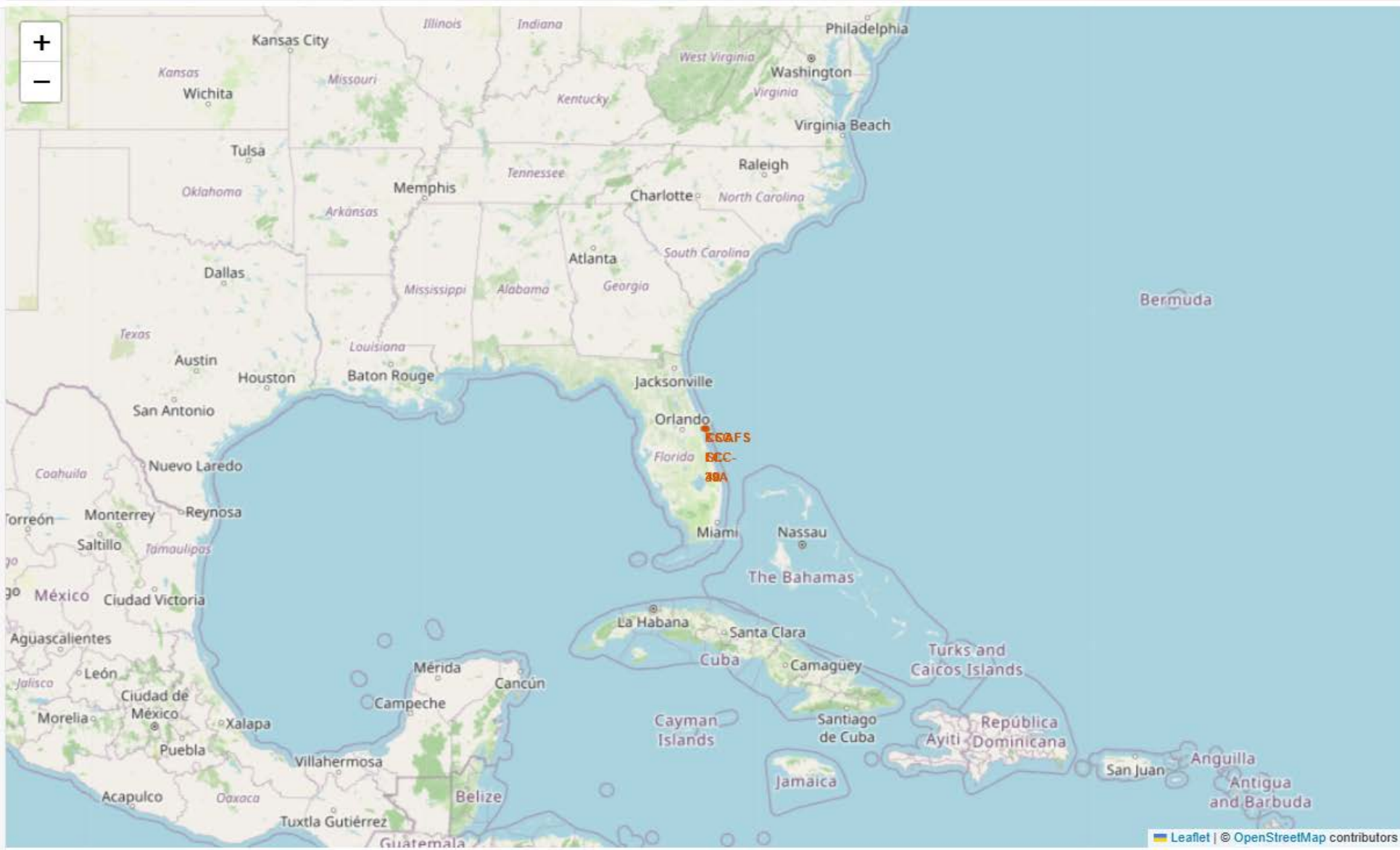
# 4. Iterate and add circles/markers
for index, row in launch_sites_df.iterrows():
    circle = folium.Circle(
        [row['Lat'], row['Long']],
        radius=1000,
        color='#d35400',
        fill=True
    ).add_child(folium.Popup(row['Launch Site']))

    marker = folium.map.Marker(
        [row['Lat'], row['Long']],
        icon=DivIcon(
            icon_size=(20, 20),
            icon_anchor=(0, 0),
            html='<div style="font-size: 12; color:#d35400;"><b>%s</b></div>' % row['Launch Site']
        )
    )

    site_map.add_child(circle)
    site_map.add_child(marker)

site_map
```

[5]:



The following dataset with the name `spacex_launch_geo.csv` is an augmented dataset with latitude and longitude added for each site.

```
[8]: URL = 'https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/datasets/spacex_launch_geo.csv'
      spacex_df = pd.read_csv(URL)
```

Now, you can take a look at what are the coordinates for each site.

```
[9]: # Select relevant sub-columns: `Launch Site`, `Lat(Latitude)`, `Long(Longitude)`, `class`
      spacex_df = spacex_df[['Launch Site', 'Lat', 'Long', 'class']]
      launch_sites_df = spacex_df.groupby(['Launch Site'], as_index=False).first()
      launch_sites_df = launch_sites_df[['Launch Site', 'Lat', 'Long']]
      launch_sites_df
```

```
[9]:
```

	Launch Site	Lat	Long
0	CCAFS LC-40	28.562302	-80.577356
1	CCAFS SLC-40	28.563197	-80.576820
2	KSC LC-39A	28.573255	-80.646895
3	VAFB SLC-4E	34.632834	-120.610745


```
[10]: # Start location is NASA Johnson Space Center
nasa_coordinate = [29.559684888503615, -95.0830971930759]
site_map = folium.Map(location=nasa_coordinate, zoom_start=10)
```

We could use `folium.Circle` to add a highlighted circle area with a text label on a specific coordinate. For example,

```
[11]: # Create a blue circle at NASA Johnson Space Center's coordinate with a popup label showing its name
circle = folium.Circle(nasa_coordinate, radius=1000, color='#d35400', fill=True).add_child(folium.Popup('NASA Johnson Space Center'))
# Create a blue circle at NASA Johnson Space Center's coordinate with a icon showing its name
marker = folium.map.Marker(
    nasa_coordinate,
    # Create an icon as a text label
    icon=DivIcon(
        icon_size=(20,20),
        icon_anchor=(0,0),
        html='<div style="font-size: 12; color:#d35400;"><b>%s</b></div>' % 'NASA JSC',
    )
)
site_map.add_child(circle)
site_map.add_child(marker)
```

Let's first create a `MarkerCluster` object

```
[12]: marker_cluster = MarkerCluster()
def assign_marker_color(launch_class):
    if launch_class == 1:
        return 'green'
    else:
        return 'red'

spacex_df['marker_color'] = spacex_df['class'].apply(assign_marker_color)
spacex_df.tail(10)
# Add marker_cluster to current site_map
site_map.add_child(marker_cluster)

# Iterate through spacex_df to add a Marker to marker_cluster
for index, record in spacex_df.iterrows():
    # Create a Marker object with custom icon color
    marker = folium.Marker(
        location=[record['Lat'], record['Long']],
        icon=folium.Icon(color='white', icon_color=record['marker_color']),
        popup=record['Launch Site']
    )
    marker_cluster.add_child(marker)

# Display the map with clustered markers
site map
```

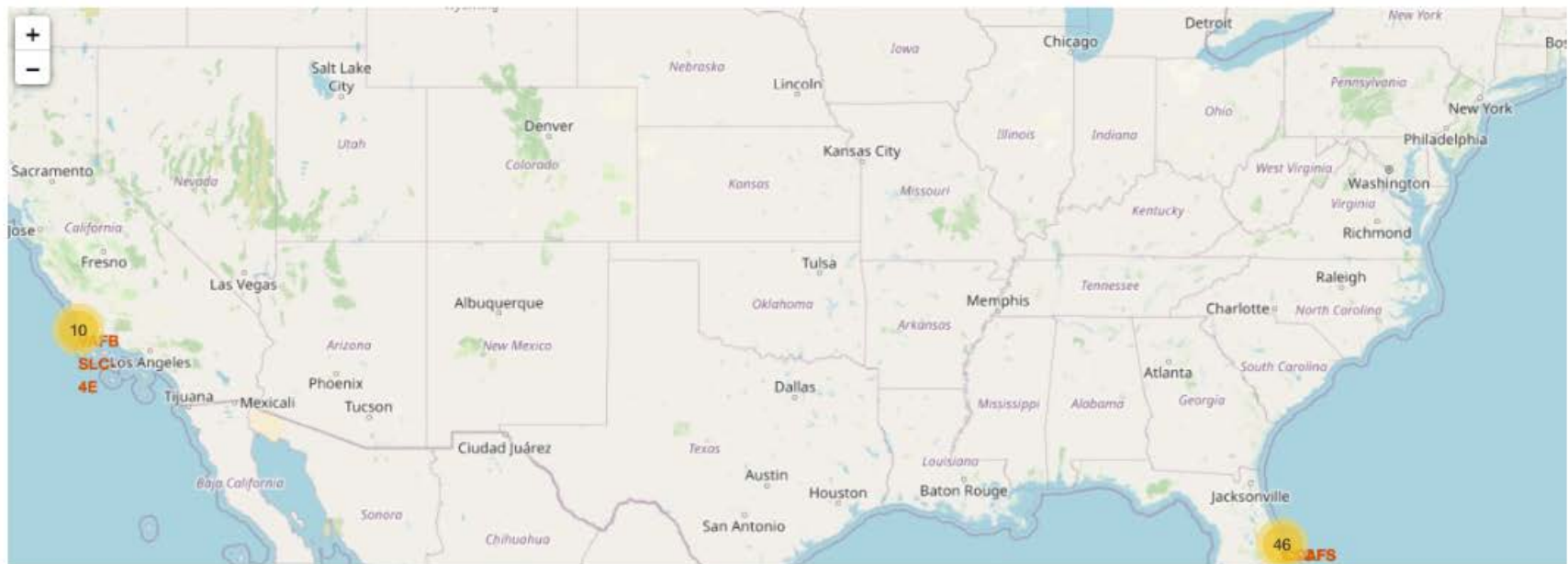
TODO: For each launch result in `spacex_df` data frame, add a `folium.Marker` to `marker_cluster`

```
[ ]: # Add marker_cluster to current site_map
site_map.add_child(marker_cluster)

# for each row in spacex_df data frame
# create a Marker object with its coordinate
# and customize the Marker's icon property to indicate if this launch was succeeded or failed,
# e.g., icon=folium.Icon(color='white', icon_color=row['marker_color'])
for index, record in spacex_df.iterrows():
    # TODO: Create and add a Marker cluster to the site map
    # marker = folium.Marker(...)
    marker_cluster.add_child(marker)
```

site_map

Your updated map may look like the following screenshots:




```
[14]: from math import sin, cos, sqrt, atan2, radians

def calculate_distance(lat1, lon1, lat2, lon2):
    # approximate radius of earth in km
    R = 6373.0

    lat1 = radians(lat1)
    lon1 = radians(lon1)
    lat2 = radians(lat2)
    lon2 = radians(lon2)

    dlon = lon2 - lon1
    dlat = lat2 - lat1

    a = sin(dlat / 2)**2 + cos(lat1) * cos(lat2) * sin(dlon / 2)**2
    c = 2 * atan2(sqrt(a), sqrt(1 - a))

    distance = R * c
    return distance
```

TODO: Mark down a point on the closest coastline using MousePosition and calculate the distance between the coastline point and the launch site.

```
[ ]: # find coordinate of the closet coastline
# e.g.,: Lat: 28.56367 Lon: -80.57163
# distance_coastline = calculate_distance(launch_site_lat, launch_site_lon, coastline_lat, coastline_lon)
```

```
[ ]: # Create and add a folium.Marker on your selected closest coastline point on the map
# Display the distance between coastline point and launch site using the icon property
# for example
# distance_marker = folium.Marker(
#     coordinate,
```